

Referee report: *The obstruction to a skew-brace bicrossed
decomposition
is a bilinear second-cohomology class*

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Title of reviewed paper: *The obstruction to a skew-brace bicrossed decomposition is a bilinear second-cohomology class* (5pp, dated 25 Sep 2026)
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Section and item numbers refer to the 5-page PDF whose front matter reads “Work-in-progress: scotmacbeth/work-in-progress @ b243b77”. All computations are in `work-in-progress/reviews/scripts/skewbrace_h2_check.py` (output in `skewbrace_h2_check.log` next to it). The script is independent of your enumerator. It enumerates every skew brace whose additive group is $\mathbb{Z}_2 \times \mathbb{Z}_4$ (28 labelled structures) or D_4 (20), and every ideal D that is a trivial brace. For each one it builds H_{sb}^2 *intrinsically*. Z^2 is the set of normalised pairs $(\beta, \bar{\tau})$ for which the twisted structure on $H \times D$ is a skew brace inducing the given (ν, μ, σ) . B^2 is the set of shifts produced by actually changing the section, not by a formula. I read Rathee–Yadav (RY) at arXiv:2601.12371v2 and their JPAA paper [20] at arXiv:2102.12235. The second one is freely available, so it can be in your corpus.

Summary of the claims

- **T1 (Def. 1, Lemma 2, Thm 3):** the additive forgetful map $\varphi_+ : [(\beta, \tau)] \mapsto [\beta]$ is well defined, because a change of section shifts β by the additive coboundary $d\theta$, with no dependence on τ . This is claimed as new over RY.
- **T2 (Thm 6):** for a trivial-brace kernel, $[\Omega] = [(\beta, \tau)]$, and $[\Omega] = 0$ iff the extension splits, iff $G \cong (G/D) \bowtie D$, iff D has a sub-skew-brace complement.
- **T3 (Thm 8, Cor. 9, Rem. 10):** on W_4 ($\mathbb{Z}_2 \times \mathbb{Z}_4$, $D = \{0\} \times \mathbb{Z}_4$), $\Phi([\Omega]) = (0, 0)$ but $[\Omega] \neq 0$. The note says this separation requires $D \not\leq \text{Soc}(G)$ and that RY Example 3.4 corroborates it.

Verdicts

1. **Your “Lemma 3.6” (in the PDF it is Lemma 2, §2): correct.** I re-derived it (below) and tested it by machine on 672 (instance, θ) pairs, including skew braces with non-abelian additive group D_4 . There were zero failures. It is also *not new*. Formula (5) is, by definition, the β -component of RY’s coboundary map ∂^1 in [20]. The “new content over RY” claim for T1 has to go.

2. **Non-injectivity of Φ (Thm 8): true.** I confirmed it by explicit cocycle/coboundary enumeration. On $\mathbb{Z}_2 \times \mathbb{Z}_4$ every (Y, Y, N) instance with a trivial-brace ideal has $|H_{Sb}^2| = 4$ and $|\ker \Phi| = 2$, with $[\Omega]$ the nonzero element of $\ker \Phi$. Below I give a hand-checkable witness. The logic of the proof of Thm 8 is sound, but W_4 is never actually written down in the note.
3. **RY identification (Thm 6, first equivalence): correct, and it is RY's.** Your transcription of RY (3.10) is right: in all 116 instances, “group cocycles + (3.10)” gives exactly the intrinsic Z_{Sb}^2 . Your quotation of RY's $\varphi_\circ$ computation is also accurate.
4. **Remark 10 and Summary T3 are false.** The separation happens with $D = \text{Soc}(G)$. RY Example 3.4 is not a separation of your kind: its additive extension does not split.

Independent derivation of Lemma 2

Let $(G, +)$ be possibly non-abelian, with D abelian and normal, and $s' = s + \theta$ where $\theta(h) \in D$. Then $-s'(h_1 + h_2) = -\theta(h_1 + h_2) - s(h_1 + h_2)$, and $-s(h_1 + h_2) + s(h_1) = \beta(h_1, h_2) - s(h_2)$ by (2). So $\beta'(h_1, h_2) = -\theta(h_1 + h_2) + \beta(h_1, h_2) + (-s(h_2) + \theta(h_1) + s(h_2)) + \theta(h_2) = -\theta(h_1 + h_2) + \beta + \mu_{h_2}\theta(h_1) + \theta(h_2)$, and all four summands lie in the abelian group D . This is your (5). The right-action convention is consistent: $\mu_h\mu_k = \mu_{k+h}$, because $s(k) + s(h) = s(k + h) + \beta(k, h)$ and conjugation by an element of D is trivial on D . Only group structure is used. τ, ν, σ never enter. I also checked the $\bar{\tau}$ -shift formula you quote from RY (p. 13 of v2, the paragraph after Prop. 3.12). It is right: $s'(h) = s(h) \circ \nu_h^{-1}\theta(h)$, and conjugating through gives $\theta_1(h_2) - \theta_1(h_1 \circ h_2) + \sigma_{h_2}\theta_1(h_1)$.

Prior art. In [20] (arXiv:2102.12235, §3, p. 6), cohomologous is defined as

“there exists a $\theta \in C_N^1$ such that $(\beta_1 - \beta_2)(h_1, h_2) = \theta(h_2) - \theta(h_1 + h_2) + \mu_{h_2}(\theta(h_1))$ and $(\tau_1 - \tau_2)(h_1, h_2) = \dots$ ”

So Thm 3 is a one-liner from the definition. The β -part of $\partial^1\theta$ is a group coboundary, so $[\beta]$ is constant on B_{Sb}^2 -cosets. RY needed an argument for $\varphi_\circ$ only because they work with τ and then renormalise to $\bar{\tau}$. Your first-principles computation is a correct check that ∂^1 matches change of section. That is worth one sentence, not a “Task 1 (crux)” section.

A hand-checkable witness for Thm 8

On $G = \mathbb{Z}_2 \times \mathbb{Z}_4$ put

$$(a, x) \circ (b, y) = (a + b, x + y + 2ab) \quad (a, b \in \{0, 1\}).$$

Then $\lambda_{(a,x)}(b, y) = (b, y + 2ab)$. This is a brace; the script asserts all the axioms. $D = \{0\} \times \mathbb{Z}_4$ is an ideal and a trivial brace.

- Additive complement: $\{0, (1, 0)\}$.
- Multiplicative complement: $\{0, (1, 1)\}$, since $(1, 1) \circ (1, 1) = (0, 1 + 1 + 2) = (0, 0)$.
- Sub-brace complement: none. It would need $g = (1, y)$ with $2y = 0$ and $2y + 2 = 0$ in $\mathbb{Z}_4$.

So the triple is (Y, Y, N) . By the logic of your Thm 8, $0 \neq [\Omega] \in \ker \Phi$. The machine agrees: $|H_{Sb}^2| = 4$, $|\ker \Phi| = 2$, $\beta \in B_+^2$, $\bar{\tau} \in B_\circ^2$, $(\beta, \bar{\tau}) \notin B_{Sb}^2$. The other labelled (Y, Y, N) case with $D = \{0\} \times \mathbb{Z}_4$ is $(a, x) \circ (b, y) = (a + b, x + y + 2ab + 2ay + 2xb)$. There $|\text{Soc}| = 2$; I suspect this is your W_4 . Separations also exist with non-abelian additive group. On D_4 there are four labelled (Y, Y, N) instances with $|D| = 4$, and in each $|H_{Sb}^2| = 2 = |\ker \Phi|$, so $\Phi \equiv 0$.

Numbered issues

Fatal (to a stated claim; the headline Thm 8 survives)

1. **Rem. 10 / Summary T3 is false.** You write:

“The separation in Theorem 8 therefore requires $D \not\leq \text{Soc}(G)$.”

and, in Summary T3, “the obstruction is bilinear and requires $D \not\leq \text{Soc}(G)$ ”. The witness above has $\text{Soc}(G) = D = \{0\} \times \mathbb{Z}_4$, since $\lambda_{(0,x)} = \text{id}$ and $(G, +)$ is abelian, and it is still (Y, Y, N) . Four of the six labelled (Y, Y, N) instances on $\mathbb{Z}_2 \times \mathbb{Z}_4$ have $D \leq \text{Soc}(G)$. The error is conflating splitting of RY’s Λ -group extension $0 \rightarrow \Lambda_D \rightarrow \Lambda_G \rightarrow \Lambda_H \rightarrow 0$ with splitting of the two component extensions. RY Thm 3.5 is about the former. I checked by brute force that in the $D \leq \text{Soc}$ cases the Λ -extension does *not* split. That is consistent with RY Thm 3.5 and irrelevant to Φ . Delete the claim, or replace it with the correct socle statement about Ψ (RY Thm 1.1/Cor. 3.8).

Fixable

2. **The RY Example 3.4 “corroboration” is wrong.** Rem. 10 says RY’s example is “consistent with, and independently corroborated by ... the mechanism is identical”. In RY’s brace $((x_1, y_1) \circ (x_2, y_2) = (x_1 + x_2, y_1 + y_2 + 2y_1x_2 + 2(x_1 + y_1)y_2), I = \mathbb{Z}_2 \times \langle 2 \rangle)$, every element outside I has additive order 4. So the *additive* extension does not split, and the triple is (Y, N, N) . I also computed $|H_{Sb}^2| = 8$ and $\ker \Phi = 0$ there. RY’s example separates Ψ (brace versus Λ -group), not Φ . I confirmed RY’s own statements: $\text{Soc} = \langle (0, 2) \rangle$, I is a trivial-brace ideal, and Λ_I has a complement in Λ_A .
3. **The novelty claim for T1 has to go.** The front matter says “The cohomological identification ... is the new proved content”, and the e-mail says “New content over RY: the ADDITIVE forgetful map ...”. As shown above, φ_+ is immediate from [20]’s definition of B_N^2 . Thm 6’s first equivalence is the RY/[20] classification (Thm 3.6 of [20]). What is left that is new is Thm 8 and whatever structure you can prove about $\ker \Phi$ (see “What is needed”).
4. **Rem. 4 draws the wrong moral.** You write:

“the coboundary map is diagonal: (3.10) couples the cocycles but not the coboundaries. This is precisely why both forgetful maps descend—and ... why their product can still fail to be injective.”

It is the other way round. The coboundaries *are* coupled, because a single θ drives both components. That coupling is the whole source of $\ker \Phi$:

$$\ker \Phi = (Z_{Sb}^2 \cap (B_+^2 \times B_o^2)) / B_{Sb}^2.$$

If B_{Sb}^2 were all of $Z_{Sb}^2 \cap (B_+^2 \times B_o^2)$, Φ would be injective, whatever (3.10) says. Corollary 9’s “lives in the coupling (3.10)” has the same problem. Your concrete sentence there (“ $\beta = d\theta_1$, $\bar{\tau} = d\theta_2$, yet no single θ ”) is the correct statement. Keep that one.

5. **W_4 is not defined.** Thm 8 is about “the skew brace with $(G, +) = \mathbb{Z}_2 \times \mathbb{Z}_4$ and $D = \{0\} \times \mathbb{Z}_4$ ”, which leaves $\circ$ unspecified. Several such braces exist, with different socles. The split triple is “imported as computed from the enumerator (scratch file)”. A reader cannot check any of it. Give $\circ$ explicitly and the three complement arguments by hand; the witness above takes four lines.

6. **Thm 6 rests on an uncitable source.** The remaining equivalences are “the registry result corrected-equivalence-1-2-3 (via Ferri, Thm 3.18), cited, not re-derived”. An internal registry entry is not a reference. “Split iff sub-skew-brace complement” is immediate: the image of a brace-hom section is a complement, and a complement gives a section. The semidirect-product description is RY Thm 4.12 (their [18]). “ $G \cong (G/D) \rtimes D$ ” as an *abstract* isomorphism is weaker than the extension splitting. State it as an isomorphism of extensions.
7. **Locator and label mismatch.** Your e-mail asks about “Lemma 3.6”. The PDF has no Lemma 3.6; the lemma is Lemma 2 in §2. Perhaps you were thinking of Thm 3.6 of RY or of [20]. Fix one or the other so future readers and registries point at the same object.

Cosmetic

8. **RY citation details.** “RY prove (their eqn. following (3.9), citing Lebed–Vendramin / Rathee–Yadav [20])”: RY cite “[20, Page 10, Eqn. (13)]”. Lebed–Vendramin is not cited there. “~line 813” and “§line 813” are pdftotext line numbers, not locators. Use “RY v2, p. 13, paragraph after Prop. 3.12”. Cite v2 (27 Jan 2026) explicitly.
9. **Justify “these actions are independent of the choice of s ”.** It holds because D is abelian (so μ is unchanged) and a trivial brace. For ν : $\lambda_{s(h)\circ d} = \lambda_{s(h)}\lambda_d$ with $\lambda_d|_D = \text{id}$. For σ : D is abelian under $\circ = +$. One line.
10. “Trust grade: proved ... airtight”. Drop the adjectives and let the checks speak.

What is needed for a publishable result

In its current form, the correct and new content is one 8-element example showing that Φ is not injective. That is a remark, not a paper. To make it a result:

1. **Describe $\ker \Phi$ structurally.** Unwinding the definitions (my derivation, not in the note) gives

$$\ker \Phi \cong C / (Z^1((H, +), D; \mu) + Z^1((H, \circ), D; \sigma)), \quad C = \{\delta \in C_N^1 : (0, \partial_\circ \delta) \in Z_{Sb}^2\}.$$

Here $\partial_\circ$ is the τ -component of RY’s ∂^1 . The reason is that $(\partial_+ \theta_1, \partial_\circ \theta_2) \equiv (0, \partial_\circ(\theta_2 - \theta_1))$ modulo B_{Sb}^2 . The script checks this formula against the direct computation in all 116 instances ($\mathbb{Z}_2 \times \mathbb{Z}_4$: 68/68, D_4 : 48/48). Proving it and computing C via (3.10) would give a genuine theorem. For example: a criterion for $\ker \Phi \neq 0$ in terms of (ν, μ, σ) , and an exact sequence $0 \rightarrow \ker \Phi \rightarrow H_{Sb}^2 \rightarrow H_+^2 \times H_\circ^2$.

2. Replace Rem. 10 with a correct socle statement, or drop it.
3. Remove the T1 novelty claim. Cite [20] for ∂^1 .
4. Do a literature check for $\ker \Phi$ beyond RY: Bardakov–Singh, Lebed–Vendramin, and the explicit H_N^2 computations at the end of §3 of [20].

Bottom line. The lemma you asked me to check is right, but it is definitional rather than new. The headline non-injectivity is right, and I can give you a cleaner witness. The socle remark and the RY-corroboration remark are wrong. Fix those, write W_4 down, and prove the $\ker \Phi$ formula. Then you have a short, correct note.